

Where there's flare, there's fair: A Battle of Weighing Judges and Fans

Summary

Dancing with the Stars (DWTS) features a persistent tension between **judges' scores** and **audience votes**. We build four connected models—**ECPC** (Elimination-Constrained Popularity Calibration), **BEPF** (Bayesian Elimination Particle Filter), **FII** (Fan Influence Index), and **EDF-TOPSIS** (Entropy-Driven Dynamic Fusion with TOPSIS)—to infer vote shares, quantify influence shifts, and redesign the fusion rule.

Fan vote estimation: With only the rules and weekly eliminations observed, we formulate vote reconstruction as an inverse inference problem under elimination constraints. ECPC infers stable relative vote-share trajectories for the percentage-combination era, while BEPF samples the posterior distribution of vote shares for rank-based formats and seasons with judge intervention. Reliability is assessed by consistency and uncertainty metrics; accuracy under “rank error ≤ 1 ” reaches **76.3%**, and posterior uncertainty increases as the format becomes more complex, especially when judge intervention is present.

Format impact and drivers: To quantify who dominates the weekly outcomes, FII uses the relative distance ratio between the final ranking and the judges' ranking versus the fans' ranking. Results indicate the percentage-combination format is most balanced (FII concentrated near 0.5), whereas the “bottom-two judge decision” shifts influence toward judges. Random-forest models further show that judges' scores are driven mainly by the **professional partner** and **age**, while fan voting is driven more by **industry background** (entertainment advantage) and **hometown region**.

Mechanism redesign: EDF-TOPSIS assigns weekly weights dynamically using entropy weighting based on distributional differences between scores and votes and then produces a composite ranking with TOPSIS. Simulations suggest it preserves suspense while reducing extreme outcomes (e.g., “popularity carry” or early elimination of high performers), improving fairness and robustness. **Sensitivity analysis supports partial vote transparency and minor intervention tuning.**

We conduct a **Monte Carlo perturbation-based sensitivity analysis**, showing that our conclusions are generally **stable** under reasonable perturbations, while greater **format complexity** increases **inference uncertainty**. Our framework is interpretable and works with limited public information, but its uncertainty rises when interventions reduce vote observability. These results provide robust support for format comparison and mechanism improvement.

Keywords: Dancing with the Stars; Bayesian particle filtering; random forest; entropy weighting; TOPSIS.

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) determines weekly eliminations and advancement by combining judges' scores with audience votes, yet the show does not release raw vote totals. This creates a natural need for quantitative study under incomplete information, reconstructing latent voting structure and evaluating how rule design shapes outcomes. [1][2]

Publicly documented rule descriptions indicate that the aggregation system has undergone notable evolution: early seasons largely relied on a rank-style combination, where judges' scores and audience votes were each converted into ordinal ranks and then merged; starting from Season 3, the rule shifted toward a percentage-based mapping, translating each component into comparable contributions that can be summed, thereby altering how the two signals influence results. [2] More recently, DWTS introduced a judges' save / intervention mechanism: when the combined result places couples in the bottom group, judges decide which couple to save, institutionally limiting extreme popularity-driven outcomes and redefining the boundary of judge–fan influence. [3][4]

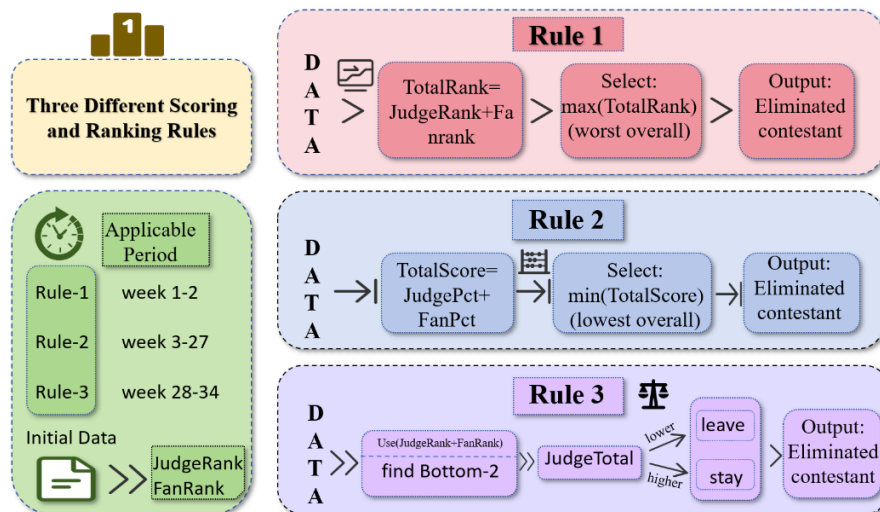


Figure 1: Judge rules

Therefore, studying the DWTS voting system is essentially a mechanism evaluation and improvement problem that must balance fairness, stability, and entertainment value—preserving public participation and suspense while avoiding over-amplification of any single signal that could undermine credibility. [1][4]

1.2 Restatement of the Problem

Considering the background information and the constraints stated in the problem statement, we aim to address the following tasks:

Problem 1. Develop a model to estimate each contestant's weekly fan voting performance during their participation. The model should produce reasonable estimates under different season rules and be validated by comparing predictions with the observed elimination records.

In addition, quantify the uncertainty of the estimates and determine whether this uncertainty varies substantially across contestants or across weeks, together with appropriate metrics.

Problem 2. Using the fan-vote estimates obtained in Problem 1, compare the show's two major historical aggregation mechanisms in terms of elimination order, advancement outcomes, and fairness, and identify which mechanism better reflects the role of fan voting. Further analyze the controversial cases provided in the prompt to explain how disagreements between judges' scores and fan votes can lead to disputed outcomes. Also discuss how introducing the "judges decide between the bottom two couples" mechanism affects outcome stability and fairness, and whether its impact is consistent across different types of contestants.

Problem 3. Build a model to evaluate the effects of the professional partner and the celebrity's own attributes on competition outcomes. The model should distinguish their roles across two channels: (i) effects on judges' scores, and (ii) effects on fan voting, and use these results to explain differences in contestants' trajectories and advancement pathways within a season.

Problem 4. Based on the models and analyses above, propose a new score-fusion/elimination mechanism, or an improvement to the current one, and justify its advantages and feasibility from the perspectives of fairness, stability, interpretability, and entertainment value. Finally, combine additional patterns or insights discovered through data mining to provide overall recommendations and conclusions for the show's producers.

1.3 Our Work

These format changes highlight the need for a principled way to quantify how judges and fans jointly shape eliminations under limited observability. To address this challenge, our work proposes an integrated modeling framework.

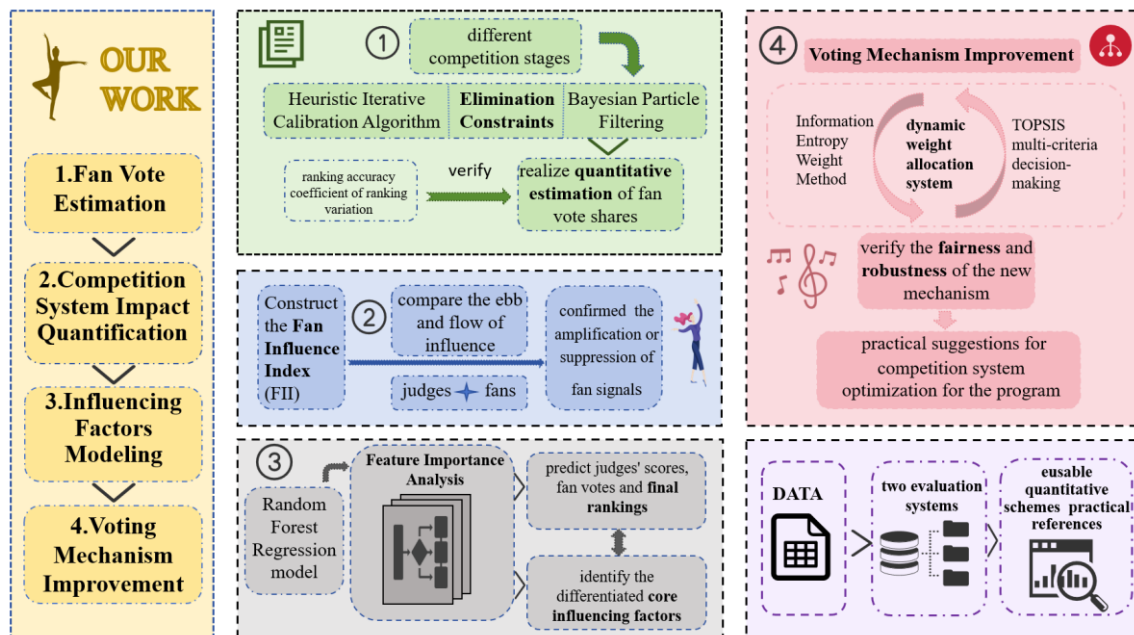


Figure 2: Our work

2 Assumptions and Notations

Assumption 1: The dataset provided in the problem is accurate. Missing values and outliers do not systematically alter the overall conclusions. When necessary, we apply consistent data-cleaning and imputation rules, and we rely on verifiable aggregated measures whenever possible during evaluation.

Assumption 2: Since raw vote counts are unavailable, we use fan support rate—i.e., a contestant's vote share—as a substitute. Within each season, a contestant's fan support rate is assumed to be approximately constant and thus applicable across different weeks in the same season.

Assumption 3: Across seasons, there may be differences in scoring scales, the number of judges, and detailed rules. We assume these differences can be absorbed by season-level features, normalization procedures (e.g., score shares or rank-based transformations), and/or model error terms, so that cross-season statistical comparisons and comparisons across rule regimes remain valid in aggregate.

Assumption 4: In stability and uncertainty analysis, we apply small random perturbations to judges' scores or vote shares to simulate realistic fluctuations. Such perturbations capture borderline-week sensitivity while reflecting only short-term noise instead of structural bias.

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Definition
$S_{i,w,k}$	Score of contestant i given by judge k in week w
$V_{i,w}$	Proportion of fan votes for contestant i in week w
$J_{i,w}$	Total score of contestant i from judges in week w
$J_{i,w}^{\%}$	Normalized percentage of contestant i 's judge score in week w
θ_i	Base popularity of contestant i for fan vote proportion
$FII_{i,s}$	Fan Influence Index: fan impact on contestant i in season s
$r_{i,w}^J$	Judge ranking of contestant i in week w
$r_{i,w}^F$	Fan vote ranking of contestant i in week w
$r_{i,w}^*$	Combined ranking of contestant i in week w
S_i (TOPSIS)	TOPSIS composite score of contestant i , used for final ranking

3 Data Preprocessing

When analyzing the dataset provided in the prompt, we adopted a unified data preprocessing procedure to ensure comparability across seasons and weeks and to avoid merge errors caused by inconsistent formats.

1. After a contestant is eliminated, all of their scores in subsequent weeks are uniformly marked as NaN (missing).

2. To support fan-vote prediction, we first compute three basic weekly indicators: the weekly total judges' score, the weekly judges' score percentage, and the weekly judges' rank. These metrics provide the data foundation for later rule-based calculations and model training.

3. For contestants who withdraw mid-season, the weeks they participated are included in rank and percentage calculations, and they are not labeled as eliminated. For finalists who have an end-of-season placement but whose elimination week cannot be determined, we also do not label them as eliminated to avoid confusion in the data.

4. When state information is missing, we fill it using the corresponding country information.

5. We standardize all string and identifier fields, including whitespace, letter case, special characters, and missing-value symbols, and we build a unified ID mapping for name matching across tables. This step ensures consistent joins across different sources and keeps downstream computations consistent, such as dancer-level aggregation for "net advantage," season-level summaries of the FII distribution, and week-level stability calculations.

After completing these steps, we generate a new CSV file with a consistent format to provide standardized input for subsequent tasks such as fan-vote estimation, rule comparison, and model training.

4 Who Gets the Fan Votes?

4.1 Statistical characterization of the raw data

The key difficulty in Problem 1 is that fan voting is confidential: the dataset does not provide raw vote counts or complete vote rankings. What we can observe are judges' scores and weekly elimination outcomes. Therefore, this problem is not well-suited to traditional regression/classification approaches that "predict votes." Instead, it should be formulated as an inverse inference problem under rule constraints: we aim to recover a plausible fan-voting structure such that, when combined with each season's aggregation rules, it can reproduce the historical elimination records.

By comparing the performance of alternative models, we adopt different inference methods for different rule regimes. In this pipeline, we estimate only the fan vote percentage (vote share), because it both reflects voting performance and can be used to derive the fan vote ranking.

Let A_w denote the set of contestants still in competition in week w . Let $S_{i,w,k}$ be the score given by judge k to contestant i in week w . Similarly, fan voting is represented as a

share $V_{i,w}$. The total judges' score for contestant i in week w is

$$J_{i,w} = \sum_{k=1}^K S_{i,w,k},$$

Within each week, we compute all quantities using only contestants who are still in the competition. We first normalize judges' scores into weekly percentages. Since the total number of audience votes is not identifiable, we do not estimate absolute vote counts; instead, we represent fan support with a season-level baseline popularity parameter and convert it into weekly vote-share percentages. The two percentages are defined as follows:

$$J_{i,w}^{\%} = \frac{J_{i,w}}{\sum_{j \in \mathcal{A}_w} J_{j,w}} \times 100\%, \quad F_{i,w}^{\%} = \frac{\theta_i}{\sum_{j \in \mathcal{A}_w} \theta_j} \times 100\%.$$

Each week, we compute statistics only for contestants in $\mathcal{A}_w$. This avoids interference from post-elimination "0-score" records and also ensures interpretability for cross-week comparisons.

4.2 Mathematical Formulation of Aggregation Rules

DWTS rules exhibit clear structural shifts over time: early seasons mainly adopted rank-based aggregation, middle seasons mainly used percentage-based aggregation, and later seasons introduced a bottom-two judges' decision mechanism. Accordingly, we divide seasons into three phases and build rule-consistent inverse-inference frameworks for each.

In Rank-based aggregation phase, we convert judges' scores and fan votes into descending ranks (a larger value implies a better rank). We then form a rank-sum measure and obtain the final combined ranking by ranking the rank-sum in ascending order, where a larger rank-sum indicates worse overall performance.

$$r_{i,w}^J = \text{rank}_{\uparrow}(\text{judge score}), \quad r_{i,w}^F = \text{rank}_{\uparrow}(\text{fan votes})$$

$$S_{i,w} = r_{i,w}^J + r_{i,w}^F, \quad r_{i,w}^{(\text{rank})} = \text{rank}_{\uparrow}(S_{i,w})$$

In Percentage-based aggregation phase. We compute each contestant's weekly combined percentage score as the sum of the normalized judges' percentage and the inferred fan vote percentage, and we rank contestants by this combined score in descending order; smaller combined scores correspond to worse performance.

$$T_{i,w} = J_{i,w}^{\%} + F_{i,w}^{\%}, \quad r_{i,w}^{*\text{pct}} = \text{rank}_{\downarrow}(T_{i,w})$$

In bottom-two judges' decision phase. We first identify the bottom-two set (two couples) using the aggregation result, and then the judges choose one of them to eliminate. To operationalize this mechanism with observable information, we use judges' performance within the bottom-two as a computable approximation, and we denote the resulting season ranking under this rule accordingly.

$$\hat{e}(w) = \arg \min_{i \in B_w} J_{i,w},$$

Handling weeks with no elimination or multiple eliminations. The dataset includes weeks with zero eliminations and weeks with multiple eliminations. To avoid incorrectly forcing a “one elimination per week” assumption, we adopt a unified bottom- k_w convention: if k_w contestants are eliminated in week w , then the model’s predicted eliminated set must fall within the bottom- k_w of the corresponding combined metric. This preserves interpretability of the rules while matching the true data structure. The distribution of weekly elimination counts (shown in the figure) illustrates why bottom- k_w is necessary rather than assuming “the single worst contestant is eliminated” each week.

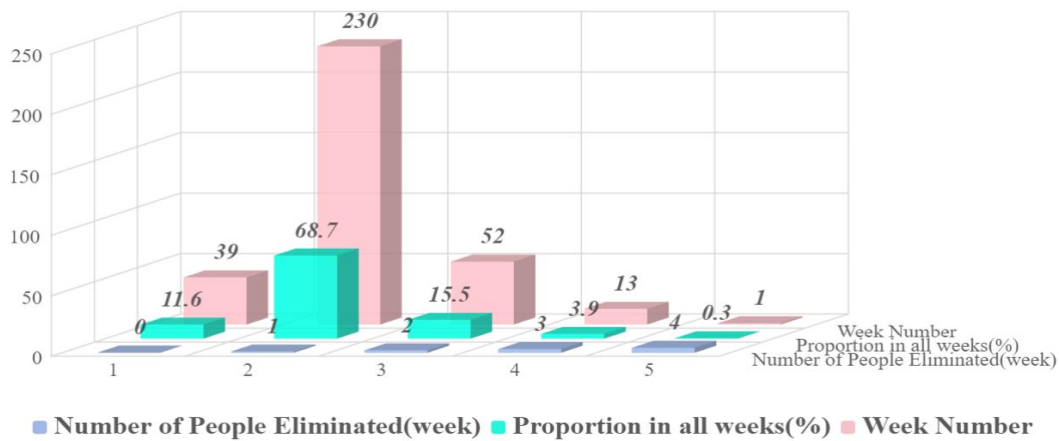


Figure 3: Distribution of the number of eliminations per week in DWTS

This distribution shows that if the model assumes “exactly one elimination per week,” it will inevitably introduce systematic errors in weeks with multiple eliminations. Therefore, in all simulations and evaluations across rule regimes, we adopt a unified bottom- k convention: when k contestants are eliminated in a given week, the model must output a set of k eliminated contestants; when $k = 0$, no elimination constraint is imposed for that week.

4.3 Iterative Calibration in the Percentage Phase (S3–27)

In the percentage-combination phase, eliminations are determined by whether a contestant falls within the weekly bottom group under the combined score. Because the fan vote percentage is unobservable, we start from a random initialization of baseline popularity parameters and generate implied vote shares, then calibrate these parameters using the observed eliminations to enforce rule consistency. If the model-implied bottom group does not include the true eliminated contestant(s), we decrease the corresponding popularity parameter(s) and renormalize the shares so that they receive less voting support in subsequent weeks. Repeating this adjustment iteratively converges toward a voting structure that can explain the historical elimination process under the published rules. Rather than claiming a unique “true” vote count, the method targets a stable, reproducible, and rule-consistent solution, which provides a unified input for the downstream analysis of fan influence.

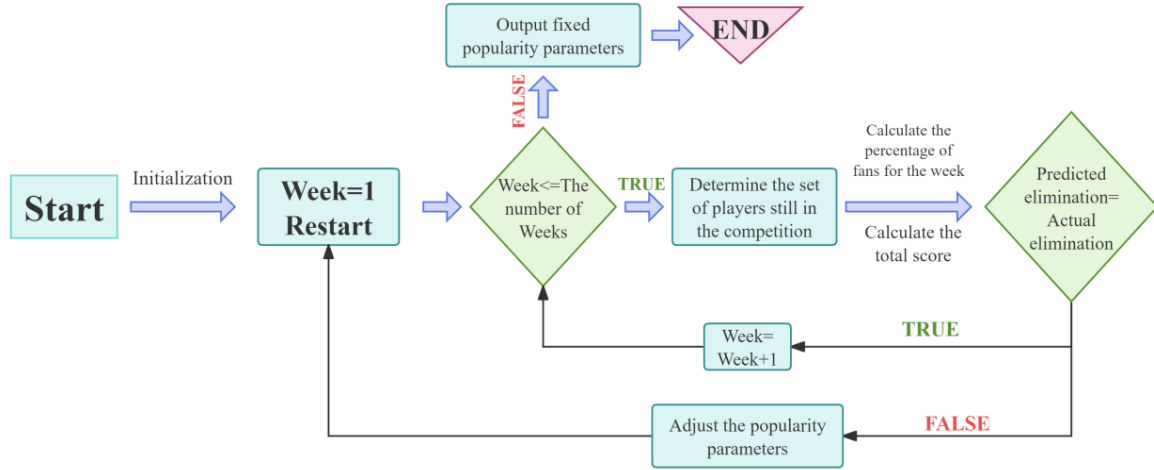


Figure 4: Iterative Flow Diagram

4.4 Bayesian Particle Filter (S1–2, S28–34)

In the rank-based and bottom-two phases, elimination observations are discrete and highly non-linear, so relying only on deterministic iteration can easily lead to local solutions. To explicitly represent uncertainty, we model the weekly fan vote-share vector with a Dirichlet prior and treat the observed eliminations as an observational constraint (likelihood), updating the posterior week by week. Because the likelihood is defined by whether the predicted eliminated set hits the observed bottom group, the posterior is difficult to derive analytically; we therefore use particle filtering for approximate inference. Each particle represents a candidate voting-structure hypothesis: particles that explain the observed eliminations receive higher weights, and resampling is performed when the effective sample size drops to prevent weight degeneracy.

Based on the Dirichlet prior and the elimination-based likelihood, we construct a particle filter that takes season-level judge information and weekly elimination records as inputs, and returns posterior samples (or a point estimate) of the fan-support parameters. The step-by-step implementation is given below.

4.5 Consistency Metrics

We evaluate model–rule consistency by focusing on contestants’ final placements, and validate whether the predicted rule mechanism aligns with the observed elimination logic by quantifying the agreement between predicted ranks and actual ranks. The main metrics include the following:

Accuracy within 1 rank: This metric measures the proportion of contestants whose absolute rank error is no greater than 1:

$$Acc_{\text{within1}} = \frac{\sum_{i=1}^N \mathbf{1}(|\hat{r}_i - r_i| \leq 1)}{\sum_{i=1}^N \mathbf{1}(r_i \neq \text{NaN})}.$$

It is the primary metric used in our implementation. By allowing small rank fluctuations,

it better reflects the subtle uncertainty around elimination boundaries in real competitions.

Mean absolute rank difference: This metric measures the average absolute deviation between predicted and actual ranks over all valid contestants:

$$\bar{\Delta r} = \frac{1}{\sum_{i=1}^N \mathbf{1}(r_i \neq \text{NaN})} \sum_{i=1}^N |\hat{r}_i - r_i| \cdot \mathbf{1}(r_i \neq \text{NaN}).$$

It summarizes the overall ranking mismatch: smaller values indicate that the predicted elimination order is closer to the observed one. In our experiments, it is computed by season and by rule regime to compare consistency across formats.

The figures visualize these consistency metrics. In the right plot formatted as $A \pm B\%$, A denotes the mean accuracy-within-1 under a given rule category, while B represents the variability across contestants; Δ denotes the mean absolute rank difference.

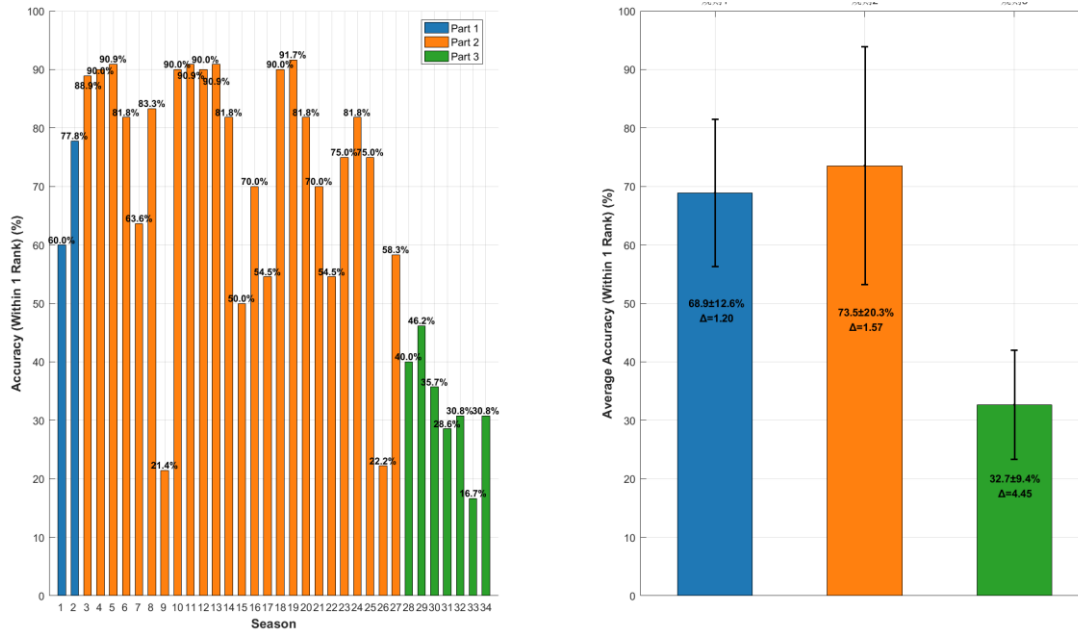


Figure 5: Consistency by season and average consistency by rule regime

4.6 Uncertainty Quantification

To quantify uncertainty in vote-share inference and verify the robustness of our conclusions, we conduct a Monte Carlo perturbation analysis based on “fan-vote perturbation + repeated simulations.” We measure certainty at both the contestant level and the season level. The core metrics and procedure are as follows.

Key metrics: Contestant-level coefficient of variation of elimination order (primary robustness metric). We measure how much a contestant’s elimination order fluctuates across repeated perturbed simulations:

$$CV_i = \frac{\text{std}(o_{i,1}, \dots, o_{i,S})}{\text{mean}(o_{i,1}, \dots, o_{i,S})}.$$

Here, $o_{i,s}$ denotes contestant i 's elimination order in the s -th simulation. A smaller CV_i indicates a more stable (robust) elimination outcome.

Season-level uncertainty index (overall certainty metric): We define a season-level uncertainty measure based on contestants' elimination confidence:

$$U_s = 1 - \frac{1}{M_s} \sum_{i \in \text{valid}_s} CS_i.$$

Here, M_s is the number of valid contestants in season s , and $CS_i \in [0,1]$ is the elimination confidence score for contestant i . A smaller U_s indicates more certain season-level predictions.

Algorithm 1. Monte Carlo Perturbation Workflow (Sensitivity Analysis)

```

1: Input: baseline fan vote shares  $F\_hat[i]$ ,
2:         weekly season data,
3:         perturbation levels  $\delta$  in  $\{0.05, 0.10, 0.20\}$ ,
4:         number of simulations  $S = 500$ ,
5:         season-specific rule type
6: Output:  $CV[i]$ ,  $SI[i]$ ,  $U_s$ , and  $Corr(\text{uncertainty, consistency})$ 
7: Step 1 (Baseline):
8:   Simulate eliminations week by week using  $F\_hat$  under the given rule
9:   Record baseline elimination order  $o\_base[i]$  and baseline predicted ranks  $r\_base[i]$ 
10: Step 2 (Perturbation simulations):
11:   for each  $\delta$  in  $\{0.05, 0.10, 0.20\}$  do
12:     for  $s = 1$  to  $S$  do
13:       Perturb vote shares and renormalize (within each active set)
14:       Re-simulate eliminations week by week under the same rule
15:       Record elimination order  $o[i, s]$  (and ranks if needed)
16:     end for
17:   end for
18: Step 3 (Metrics and screening):
19:   Compute  $CV[i]$ ,  $SI[i]$ , and season-level uncertainty  $U_s$ 
20:   Compute  $Corr(\text{uncertainty, consistency})$ 
21:   Identify the most sensitive and the most stable contestants

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The table below reports the coefficients of variation under different perturbation levels for three representative seasons selected from the three rule regimes. To further examine the season-level robustness, Fig. 6 plots how the uncertainty index and the accuracy-within-1 metric change as the perturbation magnitude increases.

Table 2: Coefficient of variation (CV) under different perturbation magnitudes

Season	perturbation5%	perturbation10%	perturbation20%
1	0.372	0.399	0.397
15	0.140	0.191	0.245
29	0.222	0.265	0.322

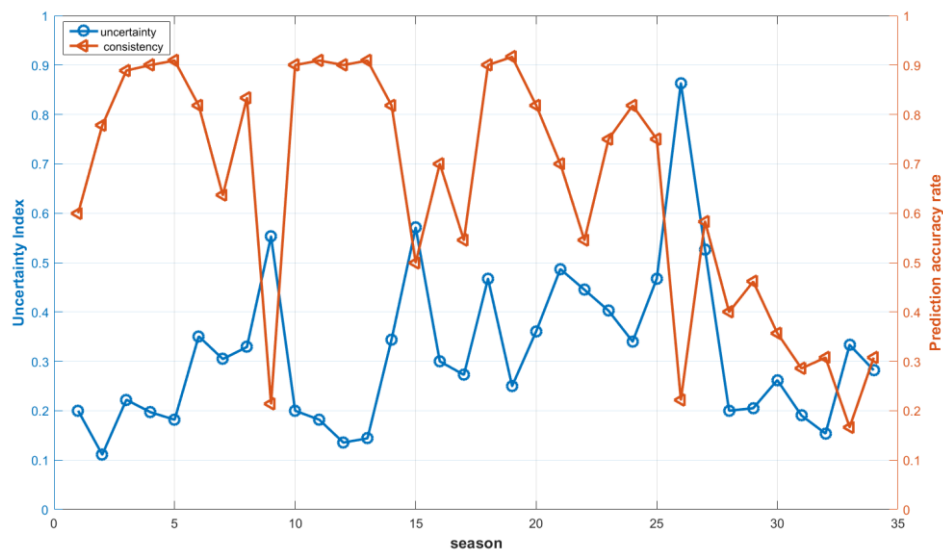


Figure 6: Changes in the season-level uncertainty index and the accuracy-within-1 metric (rank error ≤ 1) under increasing perturbation.

5 How Rules Shift Fan Influence

Within the same competition show, fan voting and judges' scoring do not combine through a single fixed linear rule. Instead, the fusion mechanism changes structurally across seasons. In some seasons, both signals are first converted into ranks and then added; in others, judges' scores are converted into percentages and combined with fan vote percentages; and in later seasons, a decisive intervention is introduced, where the bottom two couples are determined and the judges choose which one is eliminated. These rule differences mean that the same underlying distribution of votes and scores can lead to very different elimination outcomes under different formats, which in turn fuels audience narratives such as "carried by fans" or "suppressed by judges."

To quantify these differences within a unified framework, we propose the Fan Influence Index (FII) to answer a central question: in a given week's outcome, whichever side—fans or judges—produces a ranking closer to the final ranking is more likely to have dominated the elimination direction. Using this index, we compare influence patterns across rule regimes at the aggregate level and provide mechanism-based explanations for the four controversy cases

required in the prompt.

5.1 Fan Influence Index (FII)

Intuitively, if the final outcome is closer to the fans' ranking and farther from the judges' ranking, then fan voting exerts a stronger "pull" on the result. To quantify this, we define two deviation measures: the distance between the final ranking under a given rule and the judges' ranking, and the distance between the final ranking and the fans' ranking. Let $m \in \{\text{rank}, \text{pct}, \text{judge}\}$ denote the rule regime.

$$D_{i,w}^J(m) = |r_{i,w}^J - r_{i,w}^{(m)}|, \quad D_{i,w}^F(m) = |r_{i,w}^F - r_{i,w}^{(m)}|$$

We then define the Fan Influence Index (FII) as

$$FII_{i,w}(m) = \begin{cases} \frac{D_{i,w}^J(m)}{D_{i,w}^J(m) + D_{i,w}^F(m)}, & D_{i,w}^J(m) + D_{i,w}^F(m) > 0, \\ 0.5, & \text{otherwise.} \end{cases}$$

By construction, $FII \in [0,1]$ and admits a direct interpretation: values close to 1 indicate that the final outcome aligns more with fan preferences, values close to 0 indicate closer alignment with judges' preferences, and values near 0.5 suggest a roughly balanced influence between the two.

5.2 Format-Regime Comparison Results

Spearman rank correlation between rank-based and percentage-based predictions: To examine the association between the rank-based and percentage-based procedures, we visualize their agreement using a Spearman correlation heatmap. Each matrix entry is the Spearman rank correlation computed as

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2-1)}.$$

Here, n is the number of contestants in a given season-week for whom both methods produce valid predicted ranks, and d_i is the rank difference between the two methods for the i -th valid contestant. The term $\sum d_i^2$ measures the overall deviation between the two rank lists.

A value of ρ close to 1 indicates the two methods yield nearly identical weekly rankings, while $\rho = 0$ suggests no monotonic association (i.e., the two rankings behave as if unrelated).

Regime split and evaluation protocol: Following the format changes specified in the problem, we divide the show history into three regimes: S1–2 (rank-based aggregation), S3–27 (percentage-based aggregation), and S28–34 (bottom-two judges' decision).

To avoid introducing ambiguity from winners or contestants without a clear elimination week, we restrict the analysis to cases with observed eliminations and exclude withdrawals. Within each regime, we compute the Fan Influence Index (FII) under all three rule formulations

and compare their distributions.

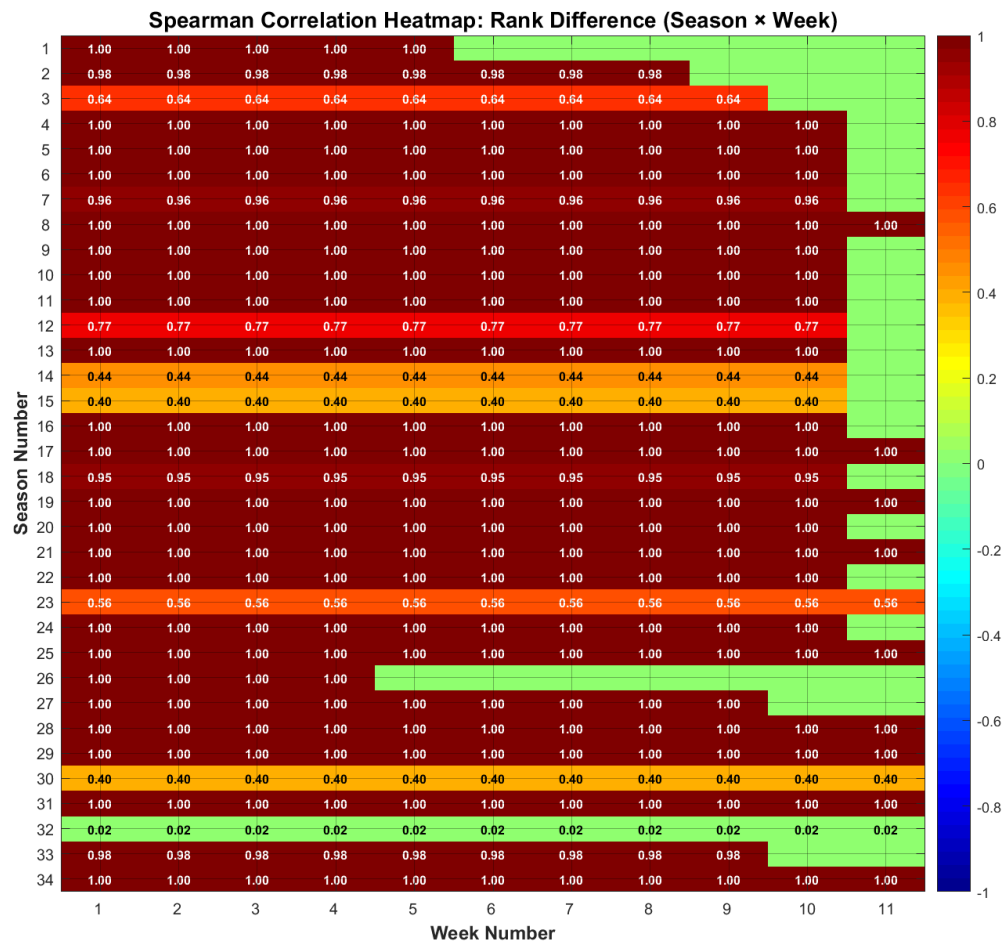


Figure 7: Spearman correlation heatmap.

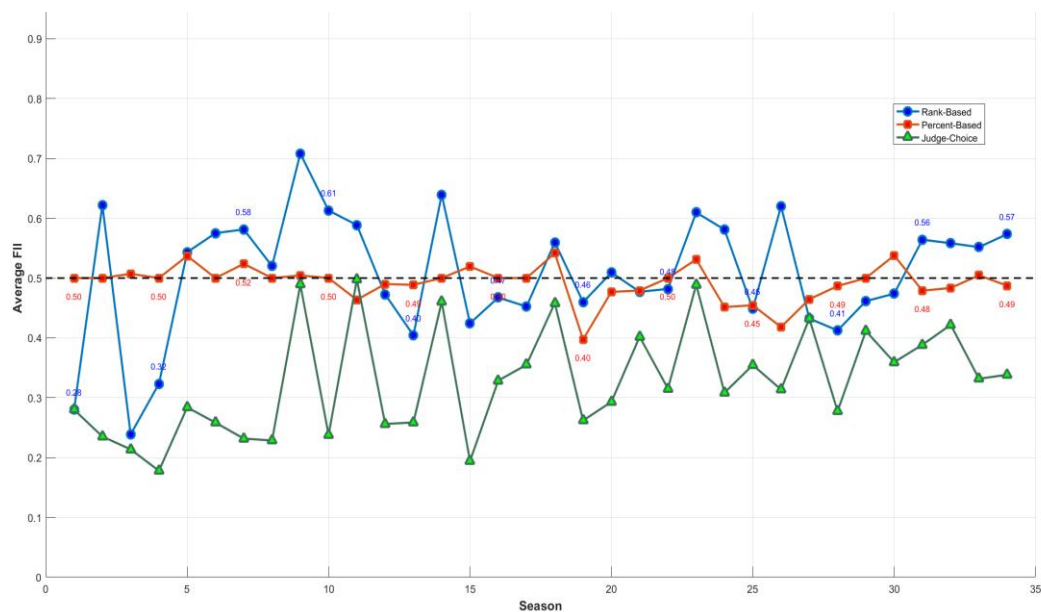


Figure 8: FII comparison across three rule formulations.

Key findings from FII comparisons: The results show that the percentage-combination rule yields FII values that are highly concentrated around 0.5, indicating the most balanced influence between fans and judges. In contrast, the bottom-two judges' decision shifts the FII distribution toward lower values overall, implying a stronger tilt toward judges.

This suggests that format rules do more than change the elimination procedure—they also reshape the extent to which fans can overturn outcomes in pivotal weeks. Once the competition is compressed into a bottom-two decision, even a strong fan-vote advantage can be effectively “cut off” by judges' intervention.

5.3 Mechanism-Based Explanations for Four Controversial Cases

The aggregate statistics show that different rule regimes shift the overall distribution of fan influence; however, the problem also asks for explanations of representative controversies. We therefore examine four iconic cases—Jerry Rice, Billy Ray Cyrus, Bristol Palin, and Bobby Bones—and compare their week-by-week FII trajectories and influence compositions under the rank-based and percentage-based formulations.

It can be seen from the data in the table that the ranking method used in Seasons 1-2 is more beneficial to fans. To reduce the influence of fans, it was improved to the percentage method. However, according to the evaluation, the percentage method only weakened the influence of fans and still had an advantage over experts. Therefore, the referee selection method was derived. The referee selection method directly grants experts the right to make decisions in the final stage, giving them an absolute advantage over fans.

Table 3: Influence composition of four controversial cases

Contestant	FII (Rank-based)	FII (Percentage-based)
Jerry Rice	53.3%	43.8%
Billy Ray Cyrus	82.9%	50.0%
Bristol Palin	81.4%	44.4%
Bobby Bones	75.6%	44.4%

6 Celebrity Effects on Outcomes

In DWTS, judges' scores primarily reflect technical quality and execution, whereas fan voting is shaped by a broader mix of signals, including dance performance, celebrity appeal, and “popularity narratives.” To quantify how strongly celebrity-related features influence outcomes—and whether they operate differently on judges versus fans—we decompose performance into three complementary target variables and model them separately: (i) a weekly baseline performance proxy that represents the judges' perspective, (ii) the fan vote-share percentage that represents audience preference, and (iii) the season-level final placement. These targets are correlated but not equivalent: final placement accumulates through the elimination process, and changing weekly weights between votes and scores can amplify or

dampen short-term advantages. Therefore, each target requires its own model.

We use random forest regression to learn the mapping from contestant features to each outcome. For a sample i , the prediction is defined as:

$$\hat{Y}_i = f(x_i, c(i)), Y \in \{R, V, P\},$$

where x_i is the feature vector and $c(i)$ is a latent grouping identifier (e.g., season membership). A random forest averages the outputs of multiple regression trees:

$$\hat{Y}_i = \frac{1}{B} \sum_{b=1}^B T_b(x_i, c(i)),$$

where B is the total number of trees and $T_b(\cdot)$ denotes the prediction function of the b -th tree. This approach avoids imposing a specific linear form and remains effective when interactions are strong (e.g., between the professional partner and celebrity attributes).

To evaluate predictive performance, we report three complementary metrics—MAE, RMSE, and R^2 —to ensure that model accuracy is assessed from multiple perspectives.

6.1 Judges' Side: Baseline Performance Prediction Model

The judges-side signal R_i is treated as a composite “starting performance” indicator. It is influenced by a contestant’s baseline ability, season-specific style effects, and the professional partner’s choreography and coaching capacity. Accordingly, we include the professional partner identity, celebrity industry, hometown state, country/region, age at participation, and season index as features, and also incorporate vote share and final placement as auxiliary joint information (consistent with our implementation). We then fit a random forest regressor to predict R_i .

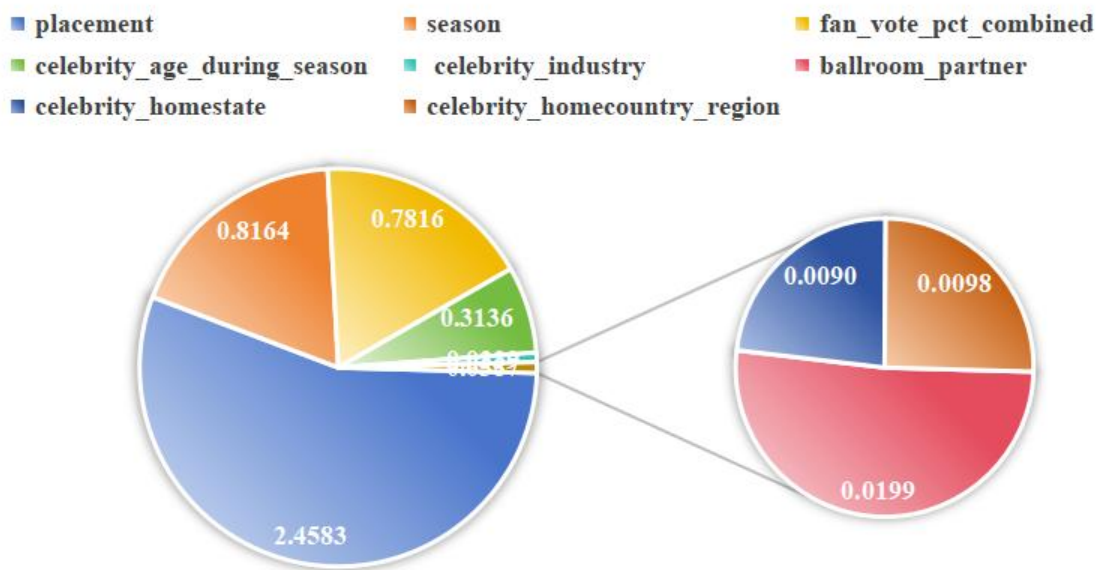


Figure 9: Feature importance for predicting judges' rank

After training, we evaluate the model using out-of-sample errors and use perturbation/permutation-style feature importance to identify which variables most strongly drive the judges-side ranking.

Table 4: Model performance

Metric	Value
MAE	2.4074
RMSM	3.1066
R	0.2834

6.2 Audience Side: Fan Vote-Share Prediction Model

The fan vote share V_i quantifies audience preference. Beyond technical performance, it can also be influenced by celebrity visibility, regional identity, and show narratives. Using the same feature set as in our implementation (professional partner, industry background, region, age, season index, final placement, etc.), we train a random forest regressor to predict vote share.

Notably, the scale and dispersion of vote shares can vary across seasons, making the season identifier an important explanatory feature in practice. To interpret what drives audience preference, we compute permutation-based feature importance. Fig.10 highlights which inputs—celebrity attributes or professional-partner labels—contribute most to predicting V_i , providing evidence for the subsequent discussion on fair fusion mechanisms.

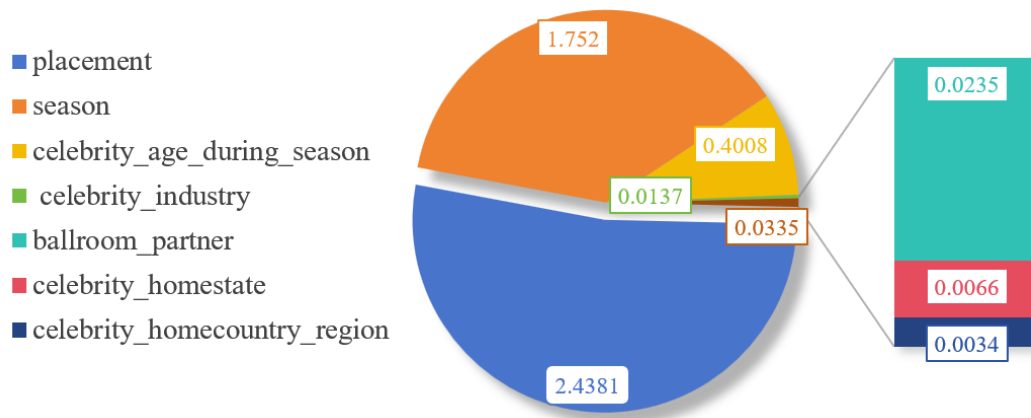


Figure 10: Feature importance for predicting fan vote share

Table 5: Model performance for fan vote-share prediction

Metric	Value
MAE	0.9976
RMSE	1.3222
R2	0.5612

6.3 Outcome Side: Final Placement Prediction Model

Final placement P_i is determined by the cumulative elimination process across the season, reflecting the long-run effects of both judges and fans. We treat P_i as a season-level outcome and model it using professional-partner and celebrity features as primary inputs, with vote share included as a key explanatory signal.

Compared with the previous two targets, final placement is more sensitive to extreme events, such as unexpected upsets or controversial eliminations in a few pivotal weeks; therefore, prediction error is typically larger. For interpretability, we again report permutation-based feature importance. Fig. Y shows the variables that most strongly explain variation in final placement.

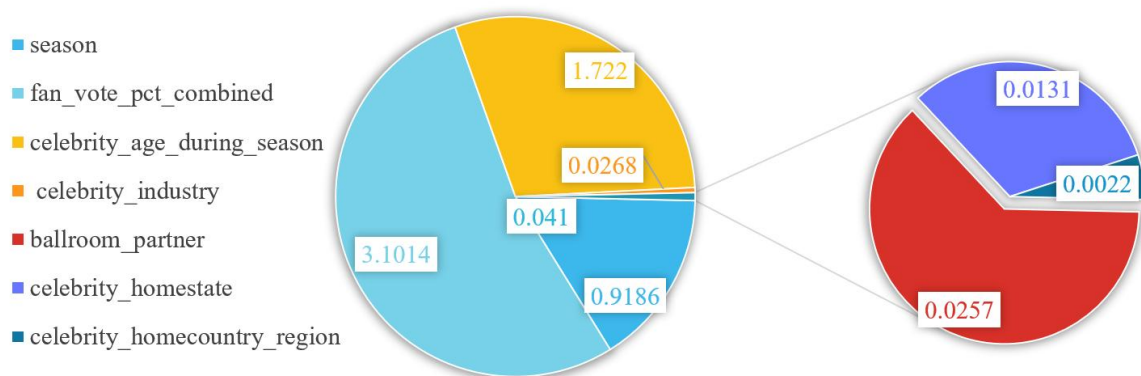


Figure 11: Feature importance for predicting final placement.

Metric	Value
MAE	1.9830
RMSE	2.6218
R2	0.5297

Table 6: Model performance for final placement prediction

7 Validation Design for the Dynamic Fairness Voting System

To verify the rationality of the proposed dynamic fairness voting system, we focus on four representative controversial contestants—Jerry Rice, Billy Ray Cyrus, Bristol Palin, and Bobby Bones. For each competition week, we compute dynamic weights via the entropy-weight method, produce a composite ranking using TOPSIS, and conduct simulation-based comparisons against historical outcomes. The key objective is to check whether the new fusion rule would cause any of these target contestants to be eliminated substantially earlier than expected, thereby ensuring that the system balances judges' expertise and audience participation while maintaining outcome fairness.

7.1 Dynamic Weight Computation via the Entropy-Weight Method

The core idea is to replace fixed weights (e.g., 50–50) with entropy-driven dynamic weights derived from the information content of judges' scores and fan votes. Under this principle, the signal providing more discriminative information in a given week receives a larger weight.

7.1.1 Normalization

Because judges' scores and fan votes may differ in scale, we first normalize the indicators using the proportion method:

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^n x_{ij}},$$

where x_{ij} is the raw value of indicator j for contestant i , n is the number of active contestants in that week, and p_{ij} is the normalized value. If an indicator has all-zero values across contestants (to avoid division by zero), we set $p_{ij} = 1/n$ for all i under that indicator.

7.1.2 Dynamic Weight Determination

Entropy measures uncertainty (or dispersion) in an indicator: a smaller entropy implies the indicator contains more effective information and better distinguishes contestants. Judges' scores are treated as a multi-week indicator (one weekly score), so we compute entropy for each week first and then average them to obtain a composite entropy for judges. Fan voting is a single indicator, so its entropy is computed directly.

For a single indicator j , entropy is defined as:

$$H_j = -\frac{1}{\ln(n)} \sum_{i=1}^n p_{ij} \ln(p_{ij}),$$

where $\ln(n)$ normalizes entropy to the range $[0, 1]$. When $p_{ij} = 0$, we set $p_{ij} \ln(p_{ij}) = 0$ to avoid undefined logarithms.

Judges' composite entropy is computed by averaging entropy values across weeks:

$$H_{\text{judge}} = \frac{1}{m} \sum_{j=1}^m H_j,$$

where m is the number of competition weeks (e.g., $m = 11$ in our setting). Fan-vote entropy is denoted by H_{fan} and is computed using the same single-indicator entropy formula.

Next, we convert entropy into a diversity (discriminatory) coefficient, where larger values indicate stronger discriminative power:

$$g_j = 1 - H_j.$$

Thus, the composite diversity coefficients for judges and fans are:

$$g_{\text{judge}} = 1 - H_{\text{judge}}, g_{\text{fan}} = 1 - H_{\text{fan}}.$$

We then normalize them to obtain the dynamic weights:

$$w_{\text{judge}} = \frac{g_{\text{judge}}}{g_{\text{judge}} + g_{\text{fan}}}, w_{\text{fan}} = \frac{g_{\text{fan}}}{g_{\text{judge}} + g_{\text{fan}}}.$$

To prevent either component from becoming negligible (and losing practical influence), we impose a minimum weight of 0.1:

$$w_{\text{judge}} = \max(w_{\text{judge}}, 0.1), w_{\text{fan}} = 1 - w_{\text{judge}},$$

$$w_{\text{fan}} = \max(w_{\text{fan}}, 0.1), w_{\text{judge}} = 1 - w_{\text{fan}}.$$

This ensures $w_{\text{judge}} + w_{\text{fan}} = 1$ while allowing influence to shift dynamically based on information contribution.

7.2 TOPSIS-Based Composite Ranking

We adopt TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) to compute a composite score and ranking for active contestants. TOPSIS ranks alternatives by comparing their distances to an ideal solution (best performance) and a negative-ideal solution (worst performance). This distance-based approach provides an objective aggregation that aligns well with the fairness objective of our voting system. The procedure is as follows.

7.2.1 Weighted Normalized Decision Matrix

We construct a decision matrix where rows correspond to active contestants and columns correspond to judges' scores (multi-week indicators) and fan vote share (single indicator). Denote

$$X = (x_{ij})_{n \times (m+1)},$$

where x_{ij} (for $j = 1, \dots, m$) is contestant i 's judges' score in week j , $x_{i,m+1}$ is contestant i 's fan vote percentage, n is the number of active contestants in the week, and m is the total number of competition weeks.

We first normalize each column using vector (Euclidean) normalization:

$$z_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^n x_{ij}^2}}.$$

If the denominator is zero (all contestants have value zero under an indicator), we set the denominator to 1 to avoid division by zero.

Next, we apply the dynamic weights to obtain the weighted normalized matrix. Let the weight vector be

$$w = \left[\frac{w_{\text{judge}}}{m}, \dots, \frac{w_{\text{judge}}}{m}, w_{\text{fan}} \right],$$

meaning the judges' total weight w_{judge} is evenly distributed across the m weekly score

columns, while fan voting receives weight w_{fan} as a single column weight. The weighted normalized entries are then

$$v_{ij} = z_{ij} w_j$$

7.2.2 Distances and Composite Score

The ideal solution consists of the best value in each column, and the negative-ideal solution consists of the worst value in each column. Since larger values are better for both judges' scores and fan vote share, we define

$$v_j^* = \max_i v_{ij}, v_j^0 = \min_i v_{ij}.$$

For each contestant i , we compute the Euclidean distance to the ideal and negative-ideal solutions:

$$d_i^* = \sqrt{\sum_{j=1}^{m+1} (v_{ij} - v_j^*)^2}, d_i^0 = \sqrt{\sum_{j=1}^{m+1} (v_{ij} - v_j^0)^2}.$$

The TOPSIS composite score is defined as the relative closeness to the negative-ideal solution:

$$S_i = \frac{d_i^0}{d_i^* + d_i^0}.$$

A larger S_i indicates better overall performance. If $d_i^* + d_i^0 = 0$ (all contestants are identical on all indicators), we set $S_i = 0.5$. If NaN values occur, we set $S_i = 0$ by convention.

7.2.3 Final Ranking

We rank active contestants in descending order of S_i . The resulting rank R_i is defined as:

$$R_i = \text{rank}(-S_i),$$

which is equivalent to sorting S_i in descending order. A smaller R_i indicates a better composite standing in that week.

7.3 Elimination Simulation and Validation

We apply the proposed fusion rule to the four controversial cases analyzed earlier and compare the simulated eliminations against the historical outcomes. Specifically, for each week, we compute entropy-based dynamic weights, obtain a TOPSIS composite ranking for active contestants, and then simulate eliminations according to the corresponding elimination rule. We then track whether the target contestants would be eliminated earlier than in the real season.

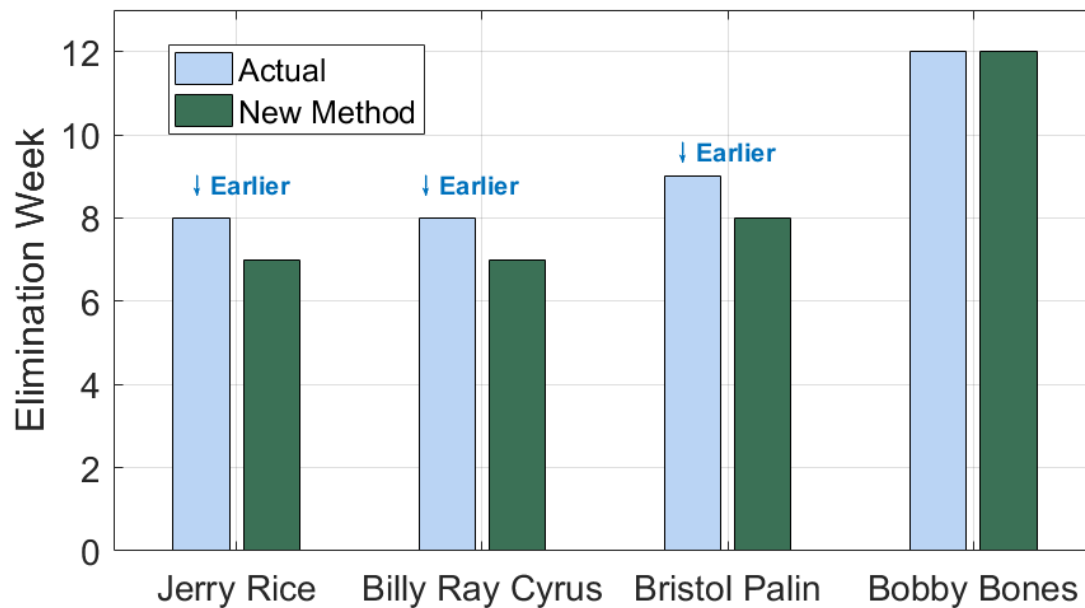


Figure 12: Elimination Week

Overall, the simulation indicates that the revised rule produces elimination trajectories that differ from the historical record and alters the elimination timing for the selected cases. This comparison-based simulation provides a direct, mechanism-level validation of how the proposed fusion system would behave under realistic weekly inputs.

7.4 Rationale and Advantages of the Proposed Method

The validation framework for our dynamic fairness voting system has two key strengths: dynamic adaptivity and objective aggregation.

Entropy-driven dynamic weighting (adaptivity): Weekly weights are computed from entropy measures rather than manually fixed ratios. By responding to the dispersion of judges' scores and audience votes, the system adjusts the relative influence of the two signals automatically, reducing the fairness bias that can arise from rigid weights.

TOPSIS-based multi-criteria ranking (objectivity): TOPSIS integrates multi-dimensional judges' information and the single-dimensional fan vote signal through distance-to-ideal comparisons, avoiding over-reliance on any single indicator (e.g., judges-only or votes-only).

Multi-faceted validation criteria (completeness): The evaluation considers multiple outcome dimensions—elimination week, ranking trajectory, and support-level changes—so it captures both contestant-level shifts and overall system-level effects, improving the reliability of conclusions.

Robust preprocessing and outlier handling (practicality): Consistent preprocessing and anomaly-handling rules improve compatibility with the raw dataset and reduce the sensitivity of validation results to data-quality issues, enhancing practical applicability.

8 Model Evaluation and Further Discussion

8.1 Strengths

1. Closed-loop inference under unobserved voting data: We reformulate fan voting as an inverse problem constrained by format rules. Using only rules and elimination outcomes, we recover a plausible relative vote-share structure and validate it via consistency with historical eliminations.¹

2. Regime-specific modeling for comparability and interpretability: We align different inference modules (calibration vs. particle filtering) with different rule regimes, avoiding a one-size-fits-all model and enabling fair, apples-to-apples comparisons of fan influence across seasons.

3. Mechanism-level, quantitative explanation of controversies: The Fan Influence Index (FII) converts “fan-dominant vs. judge-dominant” narratives into an interpretable continuous metric. Case analyses show that controversies often stem from how rules amplify or attenuate the fan signal, rather than from voting alone.

4. Actionable redesign balancing fairness and suspense: The proposed system combines entropy-based dynamic weighting with TOPSIS ranking. Simulations suggest it mitigates extreme outcomes (e.g., popularity-driven “free passes” or early elimination of strong performers) and improves fairness and robustness while preserving suspense.

8.2 Weaknesses

1. Non-identifiability limits recovery of ground truth: Without raw vote totals, our estimates represent an explainable relative structure rather than verified absolute voting truth; the framework is best suited for mechanism comparison and rule evaluation.

2. Discrete elimination signals yield higher uncertainty under complex formats: Judge intervention and irregular elimination weeks reduce informational content, leading to a more diffuse posterior and higher inference uncertainty.

8.3 Further Discussion

1. Introduce a dynamic popularity state model so fan support can evolve week by week, capturing narrative shocks and momentum shifts.

2. Incorporate external public signals (e.g., social-media engagement or search trends) to strengthen observations and further reduce vote-inference uncertainty.

9 Conclusion

This study develops an integrated framework to analyze how Dancing with the Stars format rules reshape the balance between judges’ scores and audience votes under undisclosed voting totals. By treating fan voting as an inverse inference problem constrained by eliminations, we

reconstruct plausible vote-share patterns and quantify uncertainty, then propose the Fan Influence Index (FII) to measure whether weekly outcomes align more with judges or fans. Our results show that percentage-combination formats tend to be most balanced, while bottom-two judge intervention shifts influence toward judges; controversy case studies further indicate that disputes often arise from how rules amplify or suppress the fan signal rather than from voting alone. We also model how celebrity and partner attributes drive outcomes through different channels, and propose an entropy-weighted TOPSIS fusion rule that dynamically adjusts influence and improves fairness and robustness while preserving suspense.

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When “Popularity” Meet “Professional Judgement”

How Do Judges’ Votes and Audience Votes
Jointly Shape Fairness and Dramatic Tension?

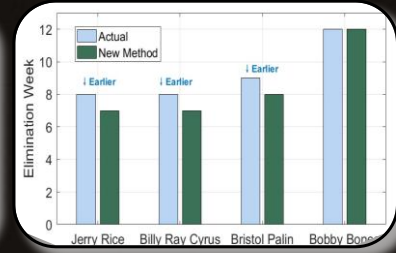
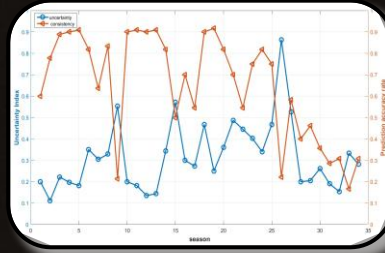


Fig. 1. Season-level relationship between reconstruction uncertainty and prediction consistency.

Blue line: inferred uncertainty index; orange line: prediction accuracy or consistency. Spikes in uncertainty are associated with sharp drops in consistency, suggesting higher explainability risk under more complex or intervention-heavy formats.

Fig. 2. Contestant-level simulation: Actual vs. New Method elimination week. For several controversial contestants, the Dynamic Weight system eliminates them earlier (≈ 1 week), aligning outcomes more closely with the dual standard of “technique + popularity,” while avoiding excessive over-correction.

I. Problems & Key Findings

Voting rules—not just performance—shape who really decides the outcome.

Our data modeling shows that the show format directly controls how much influence fans have over eliminations and final rankings. When the format is transparent and smoothly consolidated (e.g., percentage-based combination), judge and audience power tends to be more balanced. In contrast, formats that rely on rank-based aggregation and discretionary interventions can significantly compress fan dominance and amplify “rule effects,” meaning the same contestant may face very different outcomes under different mechanisms.

Uncertainty rises when the format becomes harder to infer—and accuracy drops accordingly.

Across seasons, our reconstruction uncertainty and prediction consistency move in opposite directions: when uncertainty spikes, consistency falls sharply (Fig. 1). This indicates that rule complexity and intervention-heavy structures increase the difficulty of inferring audience preferences from observable outcomes, which also increases explainability costs and controversy risk for the production team.

II. Proposed Solution: Dynamic Weight Voting Synthesis System

We propose a Dynamic Weight Voting Synthesis System that replaces fixed mixing ratios with information-driven weekly weights.

Weekly information diagnosis

The system evaluates how informative (i.e., discriminative) the judges’ scores and audience votes are each week. The signal with more concentrated and higher-discrimination information receives a higher weight.

Dynamic weighting + comprehensive ranking

After assigning weights dynamically, the system produces a final ranking using the TOPSIS method to integrate “technical quality + popularity” in a stable and explainable way.

Why this improves the mechanism:

Prevents purely popularity-driven contestants from being “locked in” by fan dominance alone.

Protects high-skill contestants from premature elimination caused by noisy or highly volatile vote swings.

Maintains suspense while improving fairness and robustness.

III. Evidence from Simulation & Practical Impact

(1) Format-level evidence (Fig. 1):

The season-level pattern shows a clear trade-off: **higher uncertainty corresponds to lower prediction consistency**, especially in seasons where rules likely introduce more non-linear effects (e.g., ranking-based aggregation or discretionary steps). This supports the claim that some formats are inherently harder to justify and communicate, even if they produce dramatic television.

(2) Contestant-level evidence (Fig. 2):

Simulations on controversial cases (e.g., Jerry Rice, Billy Ray Cyrus, Bristol Palin) show that under the new mechanism they would be eliminated **about one week earlier** than the actual timeline—indicating the system can correct over-advantage from popularity or structural artifacts. Meanwhile, for **Bobby Bones**, the new method produces an outcome consistent with the actual result, suggesting the mechanism **does not blindly overturn outcomes**, but adjusts only where the “technology + popularity” balance is most distorted.

Bottom line:

This approach offers a realistic path to preserve both professionalism and public participation. It can be piloted through **lightweight adjustments** to the existing workflow and is expected to reduce disputes driven by the rules themselves.

Report on Use of AI Tools

1. Doubao (Feb 1, 2026 version)

Query: <A brief introduction to the Bayesian particle filtering model>

Output: <Bayesian Particle Filter (PF) is a recursive state estimation method rooted in Bayesian inference and Monte Carlo sampling. Its core idea is to approximate the posterior probability distribution $p(x_t | z_{1:t})$ using a set of weighted particles $\{x_t(i), w_t(i)\}_{i=1}^N$, where $x_t(i)$ represents the state samples and $w_t(i)$ denotes the corresponding weights reflecting the matching degree between particles and observations.

The standard implementation (Bootstrap PF) follows four key recursive steps: initialization (sampling particles from prior distribution with equal weights), prediction (generating new particles via state transition model), update (adjusting weights based on observation likelihood and normalizing them), and resampling (duplicating high-weight particles and discarding low-weight ones to mitigate particle degeneracy).

Unlike Kalman Filter which is limited to linear-Gaussian systems, PF is well-suited for nonlinear and non-Gaussian state estimation tasks such as target tracking, robot localization, and signal processing. However, its computational complexity increases linearly with the number of particles, and it suffers from particle impoverishment in high-dimensional spaces.>

2. Deepseek (Jan 31, 2026 version)

Query: < What types of literature can I collect to prepare for solving this problem? Please answer briefly in English.>

Output: < To prepare for this problem, you can gather the following types of literature:

1. **TV show voting system studies** – Research on how voting and scoring systems work in competitive reality shows like *Dancing with the Stars* or *Strictly Come Dancing*.

2. **Ranking and aggregation methods** – Academic papers on rank-based vs. percentage-based voting systems, such as those in social choice theory or decision-making models.

3. **Contestant performance modeling** – Studies that analyze how factors like age, profession, or partner pairing affect audience votes and judges' scores in entertainment competitions.

4. **Audience voting behavior** – Literature on fan voting patterns, including bias, popularity effects, and engagement in televised competitions.

5. **Judging consistency and subjectivity** – Research on how judges' scores vary across performances, judges, and seasons, and methods to quantify subjectivity.

6. **Fairness in voting systems** – Papers on designing fair or exciting voting mechanisms for competitions, including hybrid systems (judge + audience input).

7. **Statistical estimation techniques** – Methods for estimating unknown data (like fan

votes) from observed outcomes, using regression, simulation, or probabilistic models.

8. ****Previous MCM/ICM papers**** – Past competition solutions related to voting, ranking, or performance prediction in similar contexts.

Focus on sources from journals in applied mathematics, statistics, social sciences, and media studies.>

3. ChatGPT (GPT-5.2 Thinking, Feb 2, 2026 version)

Query: <We asked ChatGPT to help refine and translate our report writing, including (i) compressing the abstract to fit one Word page, (ii) rewriting the first paragraph to “one-sentence background + model list,” (iii) producing concise section titles and smooth transition sentences, and (iv) formatting our methods into copy-pasteable Algorithm-style pseudocode (Particle Filter and Monte Carlo perturbation workflow) while minimizing formula symbols in running text.>

Output: <ChatGPT returned edited English and Chinese versions of the abstract/background with tighter wording and page-length control; provided multiple short slogan-style and academic-style title options; rewrote paragraphs to reduce redundancy and improve coherence; suggested which keywords to bold; generated concise section headers (e.g., shortened “How Do Format Rules Change Fans’ Influence on Eliminations?”); produced algorithm-format pseudocode for the particle filter inverse-inference procedure and the Monte Carlo perturbation sensitivity workflow; and offered wording alternatives that avoid embedding mathematical symbols in sentences to improve Word formatting and line spacing.>